CLASS 10 MATHEMATICS :: FA-4

Generating New Problems

Student Name: ..... **Roll No:** ..... **Class & Sec:** .....

1. If $\cos\theta = 4/5$, find the exact values of $\sin\theta$ and $\tan\theta$ for the acute angle θ.

2. Evaluate the numerical expression: $(\sin 30° + \tan 45°) / (1 + \cos 60°)$.

3. Prove the trigonometric identity: $(\sec\theta - \tan\theta)^2 = (1 - \sin\theta) / (1 + \sin\theta)$.

4. A tree breaks due to a storm and the broken top touches the ground making an angle of 30° with it. The distance from the foot of the tree to the point where the top touches the ground is 8 m. Calculate the total height of the tree.

5. From the top of a 75 m high lighthouse from sea level, the angles of depression of two ships are 30° and 45°. If one ship is directly behind the other on the same side, find the distance between the two ships.

6. A bag contains 5 red balls and 8 white balls. If one ball is drawn at random from the bag, find the probability that it is (i) red and (ii) white.

7. Two unbiased coins are tossed simultaneously. Determine the probability of getting at least one head.

8. A solid metallic sphere of radius 3 cm is melted and recast into small spherical balls each of radius 0.6 cm. Determine the total number of balls formed.

9. Calculate the total surface area of a solid hemisphere of radius 14 cm. (Take $\pi = 22/7$)

10. Find the volume of a right circular cone whose base radius is 6 cm and height is 7 cm.